

From Sound to Structure:

How Human Language Traverses Fragility and Time

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Introduction | Language Is Not Merely the Capacity to Speak

The origin of language has long been regarded as a tool for information transmission—an outcome of mimetic abilities (Rizzolatti & Arbib, 1998), shared intentionality (Tomasello, 2003), or even sudden linguistic module mutations (Chomsky, 2010). However, these theories often neglect a fundamental question: If imitation and social intention exist in other species, why is it that only humans developed language—and subsequently constructed fictive orders?

This study proposes that humanity's capacity for writing systems, myth-making, legal codes, identity frameworks, and knowledge institutions is not due to superior intellect or reason, but rather to our exceptional fragility and unusually long lifespan. Among all advanced mammals, only humans possess both an extended juvenile dependency and a multi-generational coexistence structure (Hrady, 2009; Kaplan et al., 2000). These biological traits generate a unique evolutionary pressure: our survival and continuity depend not on individual physical robustness but on intergenerational capacities to share risks, experiences, and intentions (Tomasello, 2008).

Building on the author's previous work, *Emotion-Driven Hypothesis for the Origins of Language* (Chan, 2025), which argues that language emerged not to describe the world but as a compensatory response to emotional misinterpretation and social prediction failure, this paper extends that hypothesis into the domain of fictive order—a high-level semantic structure that allows institutions of commitment and collective obligation to emerge from language.

We therefore propose the central thesis:

Language is a compensatory evolutionary product in response to "extended fragility."

By *extended fragility*, we refer to the long-term instability in human physiology, development, immunity, cognition, and social dependency (Sartoretti & Eberhardt, 2021; Ponsero et al., 2023). In this framework, language is not merely a sound-based communication tool, but a cognitive agency system that transcends time, roles, and events. Its functions include:

- Compressing and encapsulating non-immediate memory and experience (narrative function);
- Simulating unexperienced but potential risks (hypothetical semantics);
- Triggering collective cooperation and action (linguistic commitments, narrative norms);
- Extending into a visual-coded system (writing) to preserve long-term obligations and institutional logic (Deacon, 1997; Goody, 1986).

This work seeks to explore how language, as a tool evolved to cope with extended vulnerability, ultimately gave rise to fictive orders—such as myth, money, religion, identity, and knowledge systems—which are not byproducts, but rather the final evolutionary function of human language. Human language does more than describe reality—it simulates

unrealized structures and provides bases for collective action upon imagined futures (Harari, 2014; Dennett, 2017).

In this light, the research will be framed across several dimensions: narrative functionality, symbolic generation, grammatical progression, social control mechanisms, and the logic of writing systems. The core question shifts from "Where did language come from?" to:

How did language evolve to this point?

Chapter 1 | The Biological Peculiarity and Evolutionary Constraints of the Human Species

In many discussions on the origins of human language, one often overlooked yet crucial fact is this: humans are an exceptionally fragile and exceptionally long-lived primate species. These specific biological features demonstrate that the physiological design of humans diverges significantly from efficiency-oriented evolutionary strategies.

1.1 Longevity and Multigenerational Coexistence: An Atypical Evolutionary Choice

Compared to most primates, human lifespans are not merely long—they form a compound structure of delayed development, prolonged survivability, and multigenerational overlap (Kaplan et al., 2000). While gorillas live on average 35–40 years, and chimpanzees around 45, humans in post-agricultural societies already exhibited life expectancies of 60–80 years (Hill et al., 2007). This means that within a single human society, three or even four generations often coexist, creating a complex arena for the exchange of memory, experience, and future expectations.

1.2 Extended Fragility: The Human Infant’s Extreme Dependency

Human infants exhibit extended fragility across five major physiological indicators, especially in survivability, immunity, and threat recognition—traits significantly inferior to those of chimpanzees and gorillas. This extreme dependence suggests that imitation and real-time behavioral modules alone are insufficient to sustain human survival (Ponsero et al., 2023; Sartoretti & Eberhardt, 2021).

Physiological Domain	Human Infant	Chimpanzee/Gorilla Infant	Key Difference
Brain Development	~25% complete; requires >10 years	~40–50%; independent by age 4	Humans have a much longer delayed maturation

Independent Survival	Near zero; cannot be left for minutes	Mimics survival actions within weeks	Humans require long-term care and group support
Immune Maturity	Unstable; depends on breast milk	More rapid gut flora and immune maturity	Greater infection vulnerability, higher risk load
Food Processing Needs	Needs cooking/preparation	Can chew raw fruits and leaves	Requires cultural transfer of food knowledge
Threat Recognition	Lacks instinct; needs symbolic teaching	Learns via maternal reflex mimicry	Requires semantic instruction to avoid danger

All these indicators point to a single conclusion: Without a stable intergenerational knowledge-sharing and risk-prediction mechanism, human infants would struggle to survive beyond early childhood.

1.3 Evolutionary Pressures Toward Non-Physical Survival Strategies

Given the aforementioned physiological constraints, humans could not rely on physical explosiveness or short-term adaptation to gain evolutionary advantages. Instead, we evolved toward a unique survival package based on non-physical cognitive capacities, including:

- Long-term memory and causal inference (Schacter, 2007)
- Simulation and anticipation of others' intentions (Frith & Frith, 2006)
- Collective reliance on knowledge and emotional resonance (Hrdy, 2009)

In environments where infants are dependent for extended periods, threat detection is delayed, knowledge must be taught rather than imitated, and the causes of danger are often invisible or indirect, groups lacking language for deferred risk simulation face severe survival disadvantages. Misinterpretations and failures in intergenerational forecasting would lead to increased mortality or loss of resources.

Language, in this context, is not just an adaptive advantage—it is a biocognitive necessity, an embedded simulation module for collective memory and anticipatory reasoning.

This species-wide dependency induced a unique form of social pressure, greater than that found in any other species: we needed not only to understand each other's behaviors, but to precisely describe non-immediate events, exchange simulation models, and sustain trust across time and roles. Language became the core interface of this strategy.

1.4 Language as an Evolutionary Product of Fragility Compensation

This study proposes:

Language did not arise solely from imitation or efficiency in information transfer. It emerged because human survival needs had exceeded the cognitive bounds of immediate mimicry.

Consider the following communicative demands in early human life:

- Articulating unseen causes of illness or risk (e.g., “He got sick because he ate that fruit yesterday”)
- Constructing narrative situations that trigger empathy and cooperation (e.g., “If you were him, wouldn’t you want help?”)
- Transmitting unverifiable but actionable beliefs (e.g., “A spirit guards that mountain—don’t cut the trees”)

These needs constitute non-physiological yet existential pressures—pressures that language evolved to compensate for. As Hasson et al. (2020) argue, the neural architecture for language comprehension is not a sealed module, but an integrated predictive network combining contextual modeling, role simulation, and temporal projection. This aligns with the core claim that language evolved to simulate future risk, not merely to report present facts.

From the perspective of emergentist language models, linguistic symbols emerge gradually through the interaction of social cognition and auditory memory. When imitation-based communication fails to describe deferred events, language shifts from sound samples to abstract compositions, forming the early morphological units of verbs and causality—exactly the process hypothesized here.

In this protolinguistic form, human language ceased to be mere imitation and began functioning as a narrative simulation system—capable of compressing memory, projecting risk, and triggering shared expectations.

In the following chapter, we will analyze the core mechanisms of this simulation function, and how language operationalizes deferred risk through narrative structure.

Chapter 2 | How Language Addresses Deferred Risks and Non-Immediate Events

2.1 Narrative as a Semantic Device for Deferred Risk Management

Throughout human evolution, the most lethal threats were often not immediate, but deferred and invisible: toxic food, social ostracism, infectious diseases, or broken commitments. These events unfold across time, roles, and causality, and cannot be processed by instinct or learned through mimicry. It is precisely this triad of non-immediacy × unpredictability × collective risk that compelled the evolution of a core linguistic module: narrative capacity.

Neuroscientific studies suggest that such event simulation relies on the hippocampus for temporal memory and the prefrontal cortex for intentional modeling (Schacter, 2007). It also activates self–other switching functions (Frith & Frith, 2006). Language externalizes these internal simulations into transmissible, shareable, and reflexive semantic chains.

As proposed in *Emotion-Driven Hypothesis for the Origins of Language* (Chan, 2025), language, emotion, and intention form a self-evolving triadic structure, where:

“When emotional signals are misinterpreted and intentions are misunderstood, language becomes a necessary compensatory mechanism for social simulation.”

Narrative, then, is not a description of the world—it is a mechanism for simulating versions of the world that have not yet happened but carry behavioral consequences.

2.2 How Linguistic Symbols Encode Intergenerational Risk and Shared Intentions

Unlike memory, language has the unique ability to transcend time and social roles, enabling individuals to access experiences they did not personally encounter, yet must trust and act upon. Thus, language becomes the cognitive infrastructure for intergenerational risk management.

Take for example the utterance:

“The eastern lake once flooded our tents during heavy rain.”

This narrative contains four semantic components:

- Temporal causality: event → outcome → required action
- Risk referentiality: not happening now, but has behavioral implications
- Perspective shifting: others can inhabit the narrator’s viewpoint
- Obligation triggering: normative prescriptions for future behavior

These features cannot be achieved through gesture or mimicry alone; they require a semantic system equipped with temporality, narrativity, and hypotheticals—the very architecture that supports fictive orders.

2.3 The Limits of Non-Linguistic Memory: Comparative Animal Cases

High-intelligence social animals like chimpanzees, elephants, and parrots demonstrate memory, role tracking, and imitation. However, they cannot explain events to others, nor construct models of future collective states.

They lack the capacity to convert unverifiable memories and intentions into stable symbol chains that generate actions. As such, they cannot form institutional memory; risk remains individual and ephemeral.

Thus, language evolved not to replace perception, but to compensate for memory’s inability to cross roles and time.

2.4 From Vocal Imitation to Semantic Generation: Reconstructing the Primal Scene

To trace how language evolved from vocal mimicry to symbolic expression, we may reconstruct a plausible primal scene:

A tribe member is crushed by a falling tree and screams. A witness returns to the group and mimics the sound: “Boom!” “Aaah!” The group understands the urgency but not who is injured, how, or what to do.

Language must then perform three sequential tasks:

Stage	Description	Theoretical Function
1. Mimetic Phase	Reproducing emotional sounds	Emotional synchrony; links emotion to event perception
2. Causal Framing	Describing the mechanism (“crushed,” “broken”)	Translates raw sound into causal chains
3. Abstraction	Generating re-combinable units (“cu-”, “crack-”, “break-”)	Emergence of symbolic modularity and delayed warnings

This evolution aligns with Peirce’s triadic typology of signs:

Type	Example	Function	Evolutionary Stage
Icon	“Boom,” “Aaah”	Mimetic/emotional sync	Sensory-based imitation
Index	Pointing, gesture	Event-linked signaling	Action-sound association
Symbol	“Break,” “Danger”	Abstract representation	Intent modeling via form

Linguistic studies in sound symbolism confirm that certain sounds are non-arbitrary across languages. Examples:

Primal Sound	Likely Meaning	Cross-Linguistic Examples	Notes
<i>ca / cut</i>	cutting, rupture	<i>cut</i> (Eng), <i>kiru</i> (Jp), 砍 (Ch)	/k/, /t/ often signal force/break
<i>ma / mom</i>	maternal, feeding	<i>mom</i> (Eng), <i>ama</i> (Tib), 媽 (Ch)	universal infant sound (Jakobson)
<i>ha / hot</i>	heat, exhale	<i>hot</i> (Eng), はつ (Jp), 哈 (Ch)	voiceless exhale = thermal alert

As Roman Jakobson (1959) and later corpus studies (Blasi et al., 2016) have shown, sounds like /m/, /a/, /k/, /t/ correlate across languages with basic emotions, actions, or kinship roles. These results suggest that language did not evolve purely from social convention, but was shaped by emotional resonance and risk simulation.

The transition from emotional mimicry to modular syntax required a crisis point—what Chan (2025) refers to as the:

Expressive Collapse Threshold (ECT): The moment when existing gestural/vocal signs failed to represent emotional-intent conflict, necessitating the emergence of new symbolic structures.

2.5 Cross-Species Comparison and Semantic Pressure Equilibrium

In *Emotion-Driven Hypothesis*, the author proposed a cross-species potential matrix to evaluate why only humans developed language. Here, we extend that logic to demonstrate:

Even species capable of vocal imitation (e.g., parrots, dolphins, elephants) did not evolve syntax, likely due to the absence of expressive pressure at survival-critical thresholds.

Furthermore, in long-term human-animal interactions, the semantic burden may be absorbed by the human partner—effectively outsourcing symbolization and removing evolutionary incentive for the animal to develop its own linguistic system.

Hence, human language functions not only as a self-adaptive symbolic system, but also as a semantic exoskeleton for other species.

Summary: From Imitation to Fiction via Narrative Simulation

We may now summarize the core proposition of this chapter:

Language emerged not to state facts, but to simulate risk, compress experience, transmit expectations, and trigger trust and action.

Its evolutionary logic unfolds as:

Physiological Fragility → Memory Demand → Narrative Simulation → Predictive Modeling → Symbol Stabilization → Action Triggering

We may further delineate the inner compensatory loop as:

[Memory Demand × Emotional Misinterpretation
↓
Mutual Escalation
↑
Language as Emotional Proxy]
↓
Symbolic Evolution → Symbol Stabilization

This capacity allowed humans to preemptively act upon futures that do not yet exist, marking the threshold of civilization.

Chapter 3 | Re-Symbolization of Language — Writing as a Semantic Delay Compensation System

The original function of language was to compensate for the prediction gaps caused by extended human fragility. Evolving from vocal mimicry, narrative simulation, to fictive obligations, language enabled humans to make anticipatory decisions about events that had not yet occurred. However, once language stabilized and became responsible for institutional duties spanning across roles, generations, and situations, the immediacy and emotionality of spoken sound alone could no longer bear the weight of semantic load.

Therefore, writing did not emerge as an accessory to speech, but as a second-tier symbolic system—born from the pressure to handle *delayed semantic risks*.

3.1 The Inherent Constraints of Spoken Language: Ephemerality, Volatility, and Memory Drift

Though sound waves are highly effective in transmitting emotional resonance, they carry irreparable weaknesses:

- **Ephemerality:** Once uttered, sound disappears immediately and leaves no stable trace within temporal sequences.
- **Context Dependence:** The interpretation of meaning relies heavily on the synchrony among speaker, listener, and shared context.
- **Recall Drift:** Human memory of spoken language is unstable and highly susceptible to emotional states, expectations, and narrative reconstruction.

As human societies entered a stabilized fictive order phase, a high demand arose—not for preserving the sound itself—but for preserving the structural integrity of semantic commitments.

3.2 Writing as an Emergent Product of Pressure, Not Invention

As demonstrated in the previous chapters, language itself evolved in response to deferred risks and simulated obligations. The emergence of writing was therefore driven by pressures such as: multi-agent interactions, temporal discontinuity, and the need for semantic stability.

This was not a mere technological breakthrough, but the next stage in the evolution of symbolic systems:

- Transitioning from auditory semantic modules to visual externalization;
- Shifting from private memory proxies to public referential records;
- Converting simulated narratives into formalized obligations (e.g., legal clauses, vows, contracts).

It's important to note that the early prototypes of writing—such as oracle bone inscriptions in China or the Phoenician alphabet—were not as abstract as modern systems. Rather, they were highly pictographic in nature and emotionally evocative (Qiu, 2000; Chen Mengjia, 1963; Shirakawa Shizuka, 1971; Healey, 1990).

These visual-symbolic writing systems allowed individuals to directly associate signs with tangible objects or actions, thus lowering the learning threshold and increasing collective understanding. At this early stage of linguistic evolution, pictographic writing acted as a transitional interface between narrative simulation and institutional formalization, retaining the contextual vividness of speech while enhancing durability, reproducibility, and retrievability (Daniels & Bright, 1996).

However, writing systems in history did not always maintain this pictographic or evocative capacity. After the Phoenician script, as alphabetic systems became widespread, writing moved increasingly toward phonetic abstraction rather than semantic invocation (Coulmas, 2003). This development increased efficiency and scalability, but gradually weakened the sensorial and narrative expressiveness of written symbols.

By contrast, East Asian writing systems such as Chinese characters demonstrate a different path of development: through phonosemantic composition, radical systems, and morphemic structures, they preserve a tripartite integration of image, meaning, and sound (Qiu, 2000; Coulmas, 2003). Although they involve a steeper learning curve, such systems maintain close ties between language and sensory cognition.

This bifurcation in the evolution of writing reflects a deeper cultural trade-off: phonetic writing favors abstraction and efficiency, while pictographic writing emphasizes semantic triggering and contextual resonance (Daniels & Bright, 1996; Coulmas, 2003). Both share the same goal: to extend language beyond sensory limits and temporal discontinuities, making it capable of carrying deferred obligations, stabilizing fictive institutions, and triggering collective action.

Whether through abstract alphabets or evocative pictographs, the core function of writing is to extend the symbolic capacity of language, converting narrative and obligation into storable, transmissible, and traceable institutional structures. In this way, writing is not a mere supplement to speech—it becomes the central infrastructure through which civilization operates upon fictive order.

3.3 The Co-Evolutionary Logic of Language and Writing

We may view the emergence of language and writing as a continuous chain of compensatory pressures:

Evolutionary Stage	Core Pressure	Compensatory System	Structural Output
Emotional Misreading	Social prediction failure	Spoken language	Sensory synchrony & primary simulation
Deferred Risk	Narrative demands for non-immediate events	Narrative language	Causal chains & obligation generation
Fictive Commitment	Stability of trust and institutional memory	Written language	Contracts, laws, scriptures, records

We propose that when survival risks could no longer be mitigated through auditory forms of language alone, writing emerged to enable language to encapsulate long-distance obligations—that is:

For the first time, language became a hardware medium for institutional duties.

Although language is not biologically inherited, the pressures of obligation, narration, and institutional memory it carried gradually pushed it into a quasi-evolutionary process. This evolution did not proceed through genetic mutation or anatomical selection, but through an internal semantic self-optimization trajectory—a progressive adaptation by which language enhanced its capacity to encapsulate delayed risks and transmit institutional obligations (see also Section 2.4).

As language began to bear narrative across roles and encode institutional duties, concrete vocabulary became insufficient, prompting the system to self-expand into higher-coverage, more abstract semantic units, thereby improving narrative compression and symbolic regulation.

While the expansion of vocabulary is heavily influenced by education, social systems, and knowledge transmission, language as a cognitive proxy displays an adaptive logic similar to biological evolution. It proceeds from iconic and indexical signs to abstract symbols and hypernyms, forming a hierarchical, modular, and scalable semantic network.

Although this process may appear culturally constructed and educationally mediated, its underlying logic is not purely designed nor emergent from mutation, but rather a responsive mechanism under dynamic equilibrium. This view resonates with Rosch's (1976) theory of basic-level categories, Lakoff's (1987) exploration of semantic hierarchy and conceptual metaphor in *Women, Fire, and Dangerous Things*, Goody's (1986) work on how writing fosters institutional classification, and Deacon's (1997) thesis of language as the defining trait of a symbolic species.

3.4 A Diagram of Symbolic Hierarchy and Overview of Semantic Extension Models

We can simplify the process of linguistic re-symbolization into the following three-tier model:

1. The Sonic Symbol Layer

- Functions: emotional synchrony, crisis simulation, expressive triggering
- Examples: vocalizations, exclamations, onomatopoeia

2. The Narrative Construction Layer

- Functions: risk prediction, empathy triggering, obligation transmission
- Examples: oral mythologies, prophecies, proto-stories

3. The Written Encapsulation Layer

- Functions: externalizing obligations, institutional memory, intergenerational permanence
- Examples: religious texts, legal codes, chronicles, contracts, monetary inscriptions

This process of re-symbolization transformed language from a medium of semantic simulation into a carrier of institutional reality. It enabled language to continue functioning even across divergent temporal spans, social identities, and physical distances, maintaining its role in upholding fictive obligations and triggering action.

In other words, language was originally devised to help humans simulate unrealized futures. But once writing was introduced, it allowed humans to execute those futures—even before they happened—if they had already been prophesied, prescribed, or contractually encoded.

This marks the ultimate transformation of language as a generator of fictive order: It becomes the symbolic host of institutional structures, role obligations, and action logic.

Chapter 4 | Language as a Social Evolutionary Mechanism for Reducing Survival and Competition Costs

4.1 Reducing Survival and Competition Costs: Language as an Evolutionary Tool for Social Coordination

Throughout the highly social trajectory of human evolution, language gradually developed into a crucial tool for reducing both group survival pressures and resource competition costs. Unlike other species that rely on physical combat, intimidation, or reflexive reactions to handle conflict and resource allocation, the human linguistic system allows groups to coordinate, negotiate, and integrate resources under lower-risk conditions.

Language, in this sense, may be viewed as an evolutionary strategy for risk modulation and friction reduction, with the following concrete social functions:

- **Conflict Suppression:** Language offers a non-violent interface for communication, converting potential aggression into intentional clarification and behavioral explanation. (E.g., evolving from growling to saying, “Why did you do that?”)
- **Error Correction and Shared Information:** Narrative allows contextual background, emotional response, responsibility attribution, and even resource distribution to be negotiated at the semantic level, reducing instinctive costs.
- **Delaying Immediate Response and Enhancing Strategic Depth:** Language stretches real-time stimuli into discussable and forecastable matters, enabling reaction through simulation and social filters, resulting in deeper informational evolution and long-term knowledge transmission.

From an evolutionary perspective, this implies that language did not emerge merely for the sake of transmitting information, but to address multiple layers of pressure created by high-density social conditions—including conflict management, collaboration efficiency, and resource regulation.

The emergence of language allowed humans to shift from force- or dominance-based advantages to simulation-driven coordination. It not only reduced survival costs but also increased predictability and efficiency in collective operations.

This directly echoes the central thesis of this study:

Language emerged not from intellect or rationality, but as a compensatory strategy to mitigate the risks posed by physiological and cognitive fragility.

On a social level, it became a survival optimization module, empowering non-dominant individuals to participate in cooperation, influence decision-making, and construct action legitimacy.

4.2 The Storyteller’s Social Status and Symbolic Power: Language and the Genesis of Social Hierarchy

As language shifted from a mere risk simulation tool to a device for organizing collective order, the speaker of language gained new forms of social status. Those who could narrate,

predict, and integrate collective memory and future actions became symbolic leaders within their groups.

This narrative power did not derive from physical dominance or material possession, but from:

- **Memory Mastery:** The ability to retain and recall shared experiences and stories.
- **Narrative Construction:** The capacity to translate ambiguous or complex events into accepted group models.
- **Persuasive Resonance:** The skill to trigger emotional synchronization and behavioral consensus through storytelling.

Such linguistic competence gradually transformed into symbolic capital (Bourdieu, 1991), echoing his concept of the legitimate language—the ability to speak in ways that a group both trusts and acts upon.

In practice, those who could articulate trusted narratives held de facto authority over decision-making and resource orientation. They became agents of social design.

This process can be seen as language's first conversion into a mechanism of power—one that generates norms of expectation through narration, and establishes behavioral orders through trust.

It requires no command—only story—to establish hierarchy and institutional logic.

4.3 The Birth of Fiction: Language as Compressed Risk Simulation and Collective Action Trigger

Once language became capable of stably carrying non-immediate narratives and shared social intentions, it reached its next evolutionary phase: the emergence of fictive order.

Here, "fiction" does not mean falsehood. Rather, it refers to semantically constructed frameworks that:

- Cannot be immediately verified,
- But can trigger collective behavior.

Such fictive statements often share four core features:

- **Event Simulation:** e.g., “The ancestor spirits will punish those who break the rules.”
- **Action Directionality:** e.g., “You must not cut down this sacred tree.”
- **Role Empathy:** e.g., “If you were in their place, would you not want help?”
- **Hypothetical Consequences:** e.g., “If we don't do this, disaster may follow.”

What makes human language unique is its ability to generate belief in entities that are unverifiable yet cooperatively actionable. This semantic structure marks the transformation of risk simulation into modular narrative constructs—which are no longer descriptive, but generative.

Fiction, in this light, is a high-level narrative device evolved to compress risk, model causality, and trigger group responses. It functions not through real-time evidence, but through simulability and transmissible trust.

We may see this as the third stage of linguistic evolution:

Linguistic Stage	Function	Core Model
Stage 1: Emotional Proxy	Synchronize emotion & correct misinterpretation	Emotion–Intention Interface
Stage 2: Narrative Simulation	Encode delayed risk & shared expectations	Narrative–Simulation Matrix
Stage 3: Fictive Order	Trigger action via unverifiable models	Fictive–Action Framework

This developmental trajectory suggests that the ultimate potential of language lies in its ability to mobilize people toward unrealized worlds. It turns simulable fictions into actionable realities, and organizes belief, institutions, and obligation without relying on physical enforcement.

Language as a Pre-Institutional Simulator

This chapter argues that language is not merely a tool for sharing information or recounting experience. It is:

- A proxy system for delayed risk and emotional tension,
- A non-violent grammar for behavioral prediction and conflict regulation,
- A symbolic container for social trust and narrative authority,
- A semantic engine for triggering fictive orders and collective behavior.

At this point, language has entered the threshold of institutionalization:

- Before religion created rituals, language simulated belief.
- Before law defined obligations, language generated normative narratives.
- Before society enforced classes, language distributed roles through trust.

Fiction is not the misuse of language—it is its direction of evolution.

In the next chapter, we will analyze how language, through shared hypothetical narratives, gave rise to the five primary types of fictive orders in human society, including mythology, money, religion, law, and knowledge. We will also examine the semantic modules and trust mechanisms that support them.

Chapter 5 | The Conditions and Semantic Architecture of Fictive Order

As language evolved to handle deferred risks, simulate predictive narratives, and trigger collective action, human societies became larger, vocabularies more saturated, and the expressive load of language frequently approached a semantic saturation threshold.

At this point, a new kind of meaning structure began to emerge—one that was stable enough to drive behavior, yet not immediately verifiable. This is what we call fictive order. It is not merely a collection of imaginary stories, but rather a shared, hypothetical semantic system capable of sustaining obligation, authority, and institutional identity.

In this chapter, we examine how language evolved into a generative platform for fictive order, and unpack the semantic architecture that underpins its operation.

5.1 Language as Cognitive Proxy: Storage × Simulation × Action Mobilization

As discussed in the previous chapters, language evolved far beyond the function of exchanging information. It became a cognitive proxy—a system that performs mental tasks across time, individuals, and abstraction levels.

Its core functions include:

- **Storing Deferred Experience:** Transforming individual, bodily memory into shareable, intersubjective narrative packets;
- **Simulating Unexperienced Futures:** Generating hypothetical constructs that guide behavior, e.g., predictions of taboo violations or divine punishment;
- **Triggering Social Empathy and Norm Compliance:** Using narrative structures to evoke emotional resonance and elicit commitment.

As a cognitive proxy, language transitions from describing the known to projecting the unknown—functioning as a narrative inference engine.

This structure makes fictive order possible: a framework in which semantic structures simulate commitments, and social systems behave as if the scenario were true, even in the absence of empirical verification.

5.2 Fictive Order ≠ Fictional Narrative: Defining the Framework of Shared Hypothetical Order

While evolutionary psychology has long noted humans' attraction to fiction, this study distinguishes fictional narrative from fictive order.

As Yuval Harari (2014) emphasized, fictive orders are imagined systems that enable large-scale cooperation—like religions, nations, or money. But we propose a more precise formulation:

A fictive order is not a fanciful narrative, but a shared structure of hypothetical obligations.

This aligns with Dubourg and Baumard’s (2022) view that fiction functions as a superstimulus, activating emotional and cognitive circuits evolved for simulation and planning. Similarly, Huth et al. (2016) found that semantic categories in natural language narratives are mapped spatially across the cortex, indicating that storytelling is not just a cultural habit, but a neurosemantic mapping process.

Based on our model, a functional fictive order must fulfill the following semantic conditions:

Condition	Functional Description	Example
Deferred Verifiability	Cannot be immediately proven, but implies behavioral consequences	“Breaking ancestral rules brings disaster.”
Collective Trust Dependency	Trust relies on group consensus, not personal proof	Currency, myths, constitutions
Obligation Triggering	Narrative implies normative behavior without explicit command	“Citizens should pay taxes for protection.”
Linguistic Extendibility	Must be expandable via syntax and writing for institutional use	Laws, scriptures, knowledge systems

These conditions indicate that fictive orders are not accidents of imagination, but semantic artifacts generated by language under deferred social pressure. They are the intersections of simulation, structure, and shared belief.

Thus, language’s core capacities—empathy modeling, narrative packaging, and intergenerational encoding—are repurposed into scaffolds for imaginary yet operational worlds.

5.3 The Five Archetypal Fictive Orders: Myth, Money, Religion, Law, and Knowledge

Once language entered the fictive phase, it became a system for generating actionable frameworks that could not be verified but could still direct behavior. This use of language did not arise from fantasy, but from the need to manage unpredictable risks and non-immediate obligations under extended fragility.

In the development of human civilization, we observe five major types of fictive orders that emerged in sequence. These are not arbitrary categories; each embodies the key features of fictive structure:

- Unverifiable yet actionable
- Obligation-generating
- Narratively encapsulated and linguistically transmittable

Below is a historical projection of their likely sequence of emergence and their semantic functions:

Fictive System	Period of Initial Emergence	Core Semantic Modules	Social Function and Action Direction
Myth	Prehistoric (foraging, hunting, nomadic)	Origin stories × emotional projection × crisis modeling	Transforms invisible risks into shareable stories; early mnemonic modules (e.g., “Do not enter that forest—the spirits dwell there.”)
Religion	Late prehistory to early writing era	Taboos × reincarnation × divine mandate narratives	Systematizes myth into role systems and obligations; grants supernatural legitimacy to social order
Money	Early writing to urban stratification	Equivalence reference × trust transfer × deferred exchange	Encodes symbolic value for exchange; the first materialized fictive institution
Law	Late writing era to classical empires	Normative marking × penalty modules × language formalization	Transforms religious norms into human-authored, codified obligations; e.g., Hammurabi Code, Roman Law
Knowledge	Classical late period to early modernity	Naming logic × conceptual compression × model recomposition	Abstracts and restructures narrative into portable systems of reasoning and inference

We may also map these five fictive orders onto the trajectory of linguistic evolution:

Linguistic Evolution Phase	Corresponding Fictive Order	Core Function
Emotional Proxy Phase	Myth	Converting risks into emotionally resonant narratives
Narrative Simulation Phase	Religion	Modeling obligation through divine and role-based stories

Collective Action Trigger	Money	Converting trust into symbolic exchange and obligation
Norm Stabilization Phase	Law	Standardizing rules into explicit and codified structure
Conceptual Reassembly Phase	Knowledge	Modularizing abstraction and organizing reusability

These fictive systems are not fantasy nor misuses of language. Rather, they are the natural outgrowths of language evolving under the pressure of deferred obligation and systemic uncertainty.

If language is a compensatory mechanism for fragility,

Then fictive order is its high-level output—an advanced semantic artifact capable of encoding unreality into operational structure.

Fictive orders enable humans to act upon futures that do not yet exist, pass down accumulated memory as semantic artifacts, and transform invisible commitments into enforceable models of identity and action.

5.4 The Fusion of Fictive Orders and Semantic Recomposition Mechanisms

While the five fictive orders—myth, religion, money, law, and knowledge—can be analyzed as distinct systems in terms of origin and structure, they rarely operate independently in history. Rather, they frequently undergo semantic hybridization, nesting, and recombination.

We refer to this phenomenon as semantic complexification:

The process by which language, in modeling obligations and simulating risk, begins to merge originally distinct fictive modules into new forms of institutional logic and symbolic governance.

Below are three historically significant examples of such semantic fusion:

1. Theocracy: Myth × Religion × Law

In early civilizations, religious systems often absorbed mythic narratives, while embedding legalistic obligation. Over time, this produced religious discourse that held both narrative legitimacy and regulatory authority.

Fusion Path	Outcome / Example
Myth → Religion	Ancestral stories reinterpreted as divine mandates
Religion → Law	Taboos and rituals formalized into codified penalties
Example	The Code of Hammurabi, the Ten Commandments, Sharia, Canon Law

This kind of fusion shows how language evolved from storytelling to norm transmission, and eventually into semantic instruments of governance—where divine will becomes linguistic obligation.

2. The Emergence of Academies: Religion × Knowledge × Law

The earliest educational institutions did not originate from pure rational inquiry, but often emerged from religious training and canon interpretation. As linguistic abstraction advanced, theological language evolved into categories, logic, and scholastic frameworks.

Semantic Fusion Path	Outcome / Example
Religion × Knowledge	Theological commentary evolving into ontologies and taxonomies
Knowledge × Law	Institutional obligations and credentialing systems within academia
Example	The University of Paris, Oxford colleges, Islamic madrasas, Chinese Confucian academies

This suggests that knowledge was never purely neutral; it arose from the fusion of narrative, legal, and religious language. Its legitimacy depends not just on reason, but on the linguistic modeling of role, obligation, and epistemic structure.

3. The Birth of Insurance and Finance: Money × Myth × Knowledge

Modern insurance and financial systems appear rational, but are deeply rooted in early language-based risk simulation and fictive trust mechanisms.

Fusion Path	Outcome / Example
Myth × Money	Risk narratives transformed into value metrics
Money × Knowledge	Mathematical models (probability, statistics) encode risk as pricing
Example	Early maritime insurance, 17th-century life insurance, financial derivatives

Though expressed in numbers, financial language encodes counterfactual futures—it simulates imagined disasters or profits, underwritten by symbolic value and predictive narratives.

Conclusion: Language as a Modular System for Fictive Synthesis

These examples illustrate that fictive fusion is not just an institutional phenomenon, but a linguistic one. At each point of interaction, language functions as a:

- **Simulator:** Modeling hypothetical futures and obligations
- **Integrator:** Merging disparate semantic structures into institutional coherence
- **Packager:** Encapsulating belief, norm, and identity within transmissible narratives
- **Legitim�izer:** Rendering systems of power intelligible and acceptable via story

We can summarize this dynamic as:

[Risk Simulation] → [Semantic Naming] → [Narrative Encapsulation] →
[Institutional Encoding] → [Trust Execution]

This process enables fictive orders to evolve, hybridize, and proliferate. Language, through this mechanism, becomes not just a medium for communication, but a semantic infrastructure for organizing human belief, obligation, and cooperation.

Language does not merely create fiction—it assembles it.

Chapter 6 | Cross-Species Comparison and Integrative Modeling

Introduction | Why Did Language Emerge Only in Humans?

Language, the cornerstone of human civilization, is not merely a tool for communication. It functions as a simulation engine capable of constructing obligations, invoking trust, and organizing institutions through fictive orders. But we must confront a fundamental question:

If imitation, vocal modulation, sociality, and tool use exist in other intelligent species, why did only humans evolve language—and from it, the architecture of fictive order?

The decisive factor lies not in *capacity*, but in *pressure*.

It's not about who *can* speak, but about who *cannot not language*.

This chapter proceeds from this premise. By comparing species across biological, cognitive, social, and temporal dimensions, we propose an integrated model of language evolution that completes the theoretical framework developed thus far.

6.1 | Why Did Other Intelligent Species Not Develop Language?

As previously mentioned, species such as chimpanzees, gorillas, elephants, dolphins, and parrots share several advanced features:

- Complex social structures and role-based interactions
- Vocal imitation and non-verbal communication abilities
- Emotional recognition and some predictive behavior

Yet none of them crossed the critical threshold into a linguistic system characterized by symbolization, grammar, and narrative simulation.

The core reason is the absence of a specific configuration of evolutionary pressures.

For these species, survival and cooperation are typically resolved through immediate sensory input—gestures, calls, or physical interaction—without the need for semantic systems capable of encoding meaning across time and roles.

The critical difference is that only humans faced four interlocking pressures of deferred risk:

Dimensional Axis	Humans	Other High-Intelligence Species	Core Divergence
Physiology	Extremely fragile juveniles; require over 10 years of care	Self-sufficient by 2–3 years	Necessity to simulate future threats and long-term obligations
Generational Span	Three to four generations coexist simultaneously	Mostly single or dual generation overlap	Language must carry information and roles across generations
Social Obligation	Group actions depend on norm systems and symbolic roles	Dominance-based or reactive cooperation	Lacks role institutionalization or delayed-trust modeling
Cognitive Modeling	Possess ToM and “awareness of being observed”	Rare narrative-based role inversion or self-modeling	Lacks endogenous tension and internal simulation chain

6.2 | The Four Necessary Dimensions for Language Emergence

We synthesize these findings into what we call the “4P Matrix”—four dimensions that must simultaneously exert pressure to make language evolutionarily inevitable:

Dimension (4P)	Necessary Conditions	Functional Role
Physiology	Vocal apparatus, large brain volume, imitation ability	Hardware foundation for vocal modeling and symbol use
Psychology	Emotional complexity, intention modeling, contradiction management	Semantic logic and simulation of ambiguity
Politics (Sociality)	Role division, obligation structures, trust transference	Foundation for narrative-based cooperation
Persistence (Temporality)	Multigenerational coexistence, memory accumulation, obligation projection	Drives need for institutional memory and semantic storage

Only humans experience sustained pressure across all four axes, making language not a luxury, but a biocognitive necessity.

6.3 | Revisiting the Previous Model: Ec × Cn × Ns

In the author’s prior work *Emotion-Driven Hypothesis for the Origins of Language* (Chan, 2025), the language emergence process was modeled as an interaction between Emotional Complexity (Ec), Cognitive Architecture (Cn), and Narrative Simulation (Ns):

- Ec (Emotional Complexity): Humans process emotion across five levels (L1–L5), enabling encoding of regret, obligation, contradiction
- Cn (Cognitive Architecture): The human brain can perform imitation, abstraction, and predictive modeling simultaneously
- Ns (Narrative Simulation): Humans possess narrative reconstruction, perspective switching, and role-binding capacities

This triadic tension model constitutes the "subsurface tectonics" of language evolution. It explains why even species with vocal mimicry and high social cognition (e.g., parrots, dolphins) never developed language: they do not simultaneously endure all three internal tensions.

In other words, language did not arise from the abundance of cognitive ability, but from the depth of existential pressure.

6.4 | The Four-Stage Model of Symbolic Codification in Language Evolution

In Chapter 3, we traced the trajectory from vocal imitation to written systems. Here, we present a four-stage compensatory evolution model, showing how language progressively addressed increasing layers of deferred risk and obligation:

Stage	Primary Pressure	Compensatory Module	Structural Output
1: Emotional Proxy	Misinterpretation of feelings	Vocal synchrony & mimicry	Sound-symbol alignment (e.g., "Boom", "Aaah")
2: Narrative Simulation	Deferred risk and memory compression	Perspective-based storytelling	Role-switching, cause-effect modeling
3: Fictive Obligation	Need to stabilize collective norms	Shared belief constructs	Fictive orders: myth, taboo, divine mandates

4: Symbolic Formalization	Institutional continuity across time	Writing systems, laws, contracts	Codified language: alphabet, law, religious scripture
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Each functional layer of language corresponds to an evolutionary tension embedded within biology and social structure—none of which could be bypassed in human development.

This progressive codification reflects language’s increasing load-bearing role in transmitting trust, risk models, and institutional duties.

Just as evolution builds bones to bear heavier weight, symbolic structures were evolved to carry semantic obligations across time, space, and generations.

Summary | Language: A Threshold Emerged at the Intersection of Pressures

The emergence of language in humans—unlike in any other species—was not the result of innate superiority or cognitive excellence. While we cannot dismiss certain biological advantages in human anatomy (such as vocal tract flexibility or soft palate precision), it is the intersection of survival pressures that demanded compensatory mechanisms which ultimately led to language.

The compensatory structure proposed in this chapter, together with the previously introduced $Ec \times Cn \times Ns$ model, can be understood as a dual-axis system:

- The $Ec \times Cn \times Ns$ model explains the psychological motivational chain underlying the emergence of language;
- The four compensatory stages presented here map the institutional trajectory and functional expansion of language.

If future research can apply this dual-axis framework to neural structures, cross-species behavior, and language generation strategies, it may pave the way for a new stage in language origin studies—one that is both operationalizable and predictive across domains.

In the next chapter, we will shift our focus to a new philosophical and institutional question:

- Has this compensatory system of language now generated a new form of fragility in return?
- Has the over-symbolization of language led humanity into a deeper fracture in understanding and connection?

Chapter 7 | Language as a Compensatory Project of Consciousness

7.1 A Non-Mutational Origin: Language as an Adaptive Response to Intersecting Pressures

In the preceding six chapters, we have progressively constructed a model of compensatory evolution: extended fragility → narrative simulation → fictive order → institutional encapsulation → symbolic re-codification. Each stage was not a linear design but a structural response to different layers of “unbearable lightness”—namely: the inability to predict, remember, empathize, or trust.

Language did not emerge from a singular genetic mutation, but from the long-term convergence of physiological, social, cognitive, and temporal pressures.

As Hoemann, Gendron, and Barrett (2020) note, language is not merely a descriptive tool—it is an active mechanism for constructing emotions and expectations. Likewise, Friston et al. (2017) have positioned language as a core agent within active inference systems. In this light, the question is not “how did language arise?” but rather:

Why could language not have *not* arisen?

7.2 Fictive Capacity Is Not a Byproduct but the Ultimate Social Function of Language

From vocal imitation to divine punishment, human language has never merely described reality—it has served as a tool for transmitting knowledge, simulating scenarios, encoding obligations, and evoking trust. As Dubourg and Baumard (2022) propose, the human preference for fictional narratives stems from their ability to activate the brain’s reward circuitry for social simulation and future exploration.

We argue that the endpoint of language evolution is not a more efficient descriptive system, but a fiction-generating engine that enables humans to act within unrealized worlds.

Legal systems, religions, monetary systems, and knowledge traditions are not linguistic malfunctions—they are the high-order outputs of language’s social design.

7.3 Does Excessive Symbolization Introduce New Layers of Fragility?

Language once served as a buffer for our biological vulnerabilities. Yet, once institutionalized through codes, formulas, classifications, and databases, it risked turning its trust mechanisms into rigid hierarchies and semantic distortions.

Modern society reveals this clearly: from educational elitism and legal opacity to informational asymmetry in commerce, language’s excessive codification has pushed the threshold of access and comprehension to increasingly exclusive levels. We must ask:

Are we truly advancing with our linguistic systems—or merely outsourcing understanding to opaque structures?

As Hutchins (2021) warns, when language and thought are increasingly mediated by algorithms, humans begin to offload their cognitive labor to machines and corpora. This

transition risks reducing language from a tool of intention and meaning into a surface code for pattern manipulation.

Illich (1973) and Bauman (2000) long cautioned that tools and institutions, when over-relied upon, may betray their original purpose. The same applies to language: as it drifts from lived experience and emotional grounding into a self-replicating symbolic loop, its narrative and predictive powers risk being eroded by efficiency and standardization.

7.4 Language Re-Mediation and the Cognitive Debt of Modernity

Returning to the concept of the “Expressive Collapse Threshold (ECT)” introduced in *Emotion-Driven Hypothesis for the Origins of Language* (Chan, 2025), we recall that language first emerged when existing vocal or gestural codes could no longer bear the load of emotional-intent conflict. Language and writing thus became a breakthrough platform for reconstructing expression—allowing humans to simulate future obligations, deferred risks, and unrealized roles.

However, in today’s AI-mediated language environment—especially with the rise of large language models capable of mass generation and predictive responses—the role of language is shifting again. We no longer rely solely on our internal semantic agency to construct narratives; we can now summon pre-trained syntax and templates on demand.

This creates a phenomenon we call semantic outsourcing: while lowering the barrier to expression, it also generates a new fragility—cognitive debt.

A 2025 experiment by Kosmyna et al., *Your Brain on ChatGPT*, found that co-writing with LLMs led to declines in neural connectivity, memory integration, and a loss of language ownership. The findings suggest that when language is no longer shaped by internal tension but externally retrieved, its experiential dimension collapses.

We are thus witnessing a modern *Expressive Collapse*—not born of inner pressure but of outer convenience. Language is no longer forced to evolve through expressive failure, but instead loses generative conditions through disuse. It becomes not a survival scaffold, but an interface of passive retrieval.

This transition marks a troubling inversion: the original ECT was a site of innovation, where linguistic failure gave birth to new semantic forms. Today’s collapse, by contrast, is one of decline without reinvention—functionally extended but internally eroded.

In such a world, are we still speaking a language—or merely borrowing its skin?

7.5 Looking Ahead: Verifiability of Fictive Capacity and AI Simulation Challenges

Can the fictive-generating capacity of language be scientifically verified? Can AI truly simulate semantic cognition? These are the pivotal questions of next-generation interdisciplinary research.

Possible Indicators for Fictive Simulation:

- Theory-of-Mind activation (Frith & Frith, 2006)
- Deferred causal reasoning tasks (Lake et al., 2017)
- Neural resonance during narrative construction (Dubourg & Baumard, 2022)

AI's Current Bottlenecks:

As Bender & Koller (2020) and Mitchell et al. (2021) argue, LLMs can produce grammatically correct sentences, but do not *understand* them—they lack emotion, intent, and the capacity for obligation-role modeling.

Luciano Floridi (2014) suggests that language in the infosphere is becoming more like protocol than narrative—structured for interaction, not introspection. This shift demands both caution and creative counter-design.

Recent work by Lu et al. (2025) on EEG2TEXT-CN demonstrated a modest increase (from 0% to 6%) in aligning Chinese EEG patterns with generated text using contrastive learning and LLMs. While promising in technical fusion, this model still relies primarily on image-noun correlation and lacks the narrative, obligation, and role-based structures necessary for full semantic comprehension.

To align more closely with the semantic nature of Chinese—or of human language itself—future models may need:

- A “visual–noun–neural” grounding corpus
- Integration of narrative simulation tasks
- Context-sensitive pragmatic inference mechanisms

These steps could push AI toward a more authentic alignment with language as a simulator of deferred risk and fictive intent.

Summary | Language: Born of Pressure, Carried into Fiction, Ensnared in Codes, Awaiting Renewal

Language is the mind's answer to its own fragility—a cognitive machine born of necessity, capable of encoding risk, simulating futures, and transmitting obligation.

It has allowed us to overcome our limitations through simulation, yet now risks becoming a self-replicating shell divorced from emotional tension and narrative depth.

As Dennett (1991) once said:

“Consciousness is the brain's fictional narrator.”

Then language is the narrator's pen.

We must ask:

Will the next mutation of language be phonetic, symbolic, algorithmic—or will it be a redesign of the very kind of fiction humans can believe in?

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Chapter 9 | Postscript

In the beginning, I was haunted by a deceptively simple question:

Why did only humans develop language?

That question led me to complete *Emotion-Driven Hypothesis for the Origins of Language* (EDH), in which I attempted to explain the emergence of language as a necessary response to three converging pressures: emotional tension, cognitive simulation, and narrative demand.

But after completing that first work, I realized the question remained unresolved.

While EDH outlined the conditions for language to emerge, another central feature of human language—**symbolization**—still lacked a clear explanation. In particular, how language transcended immediate expression to become written, stabilized, and capable of carrying knowledge, systems, and institutions.

So I began writing this second piece. It originally set out to explore the question:

How does language enter the domain of fiction and institutional structure?

But partway through the writing, I came to a realization: the concept of the **Expressive Collapse Threshold (ECT)** did not just underpin the origin of language. It also echoed throughout its evolution, its socialization, and its transformation into symbolic order.

These two papers—EDH and the present work—thus form a kind of **bidirectional structure**. EDH proposes the initial pressure matrix for language emergence; this paper extends that logic, showing how language responds to those pressures by evolving into fiction, symbols, and institutional frameworks. Together, they represent my most intuitive and original response to the question of what language truly is.

Both papers were originally written in Chinese. As someone working outside the formal academic system, it has been a significant challenge to articulate these ideas fully—and to ensure their meaning is preserved in English translation. The English versions were translated with the assistance of AI (GPT), and I ask for your understanding if any phrasing feels awkward or unnatural.

If you have thoughts, critiques, or related questions, I warmly welcome you to reach out. I hope these explorations can offer something of value to your own inquiries—into language, into fiction, into structure, and perhaps into consciousness itself.

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Regards,
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Taiwan.